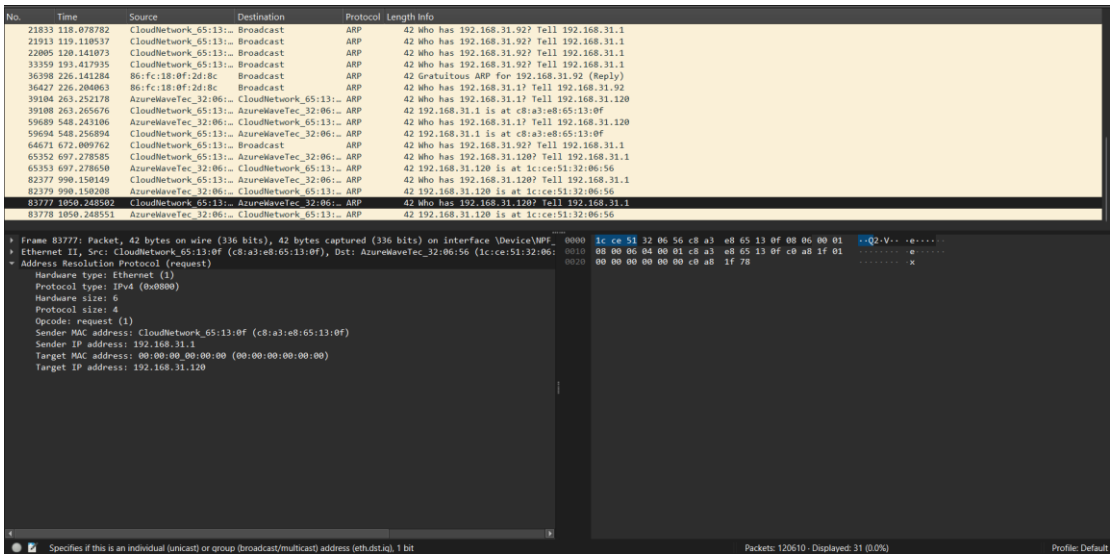
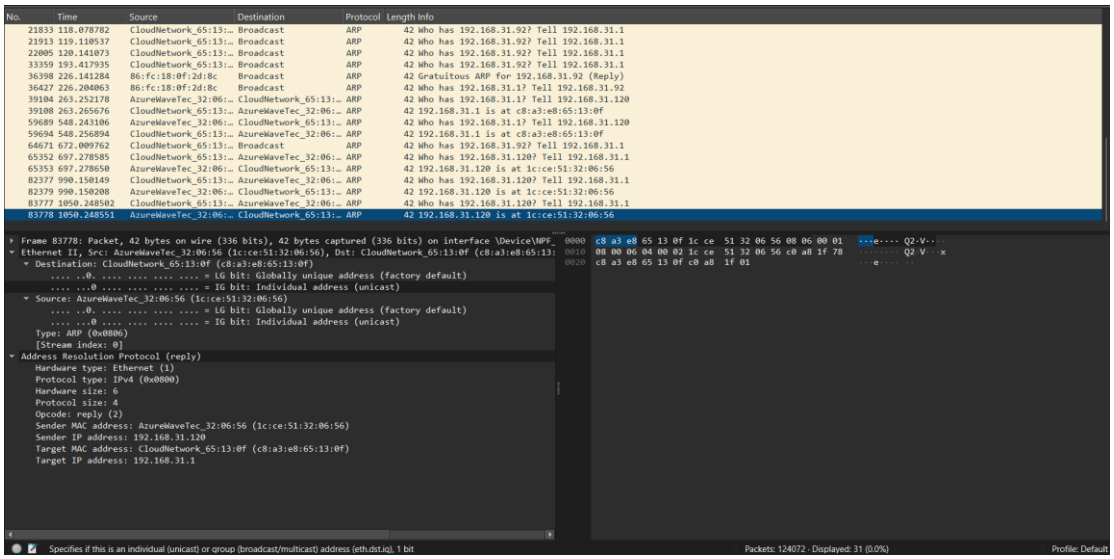
1)

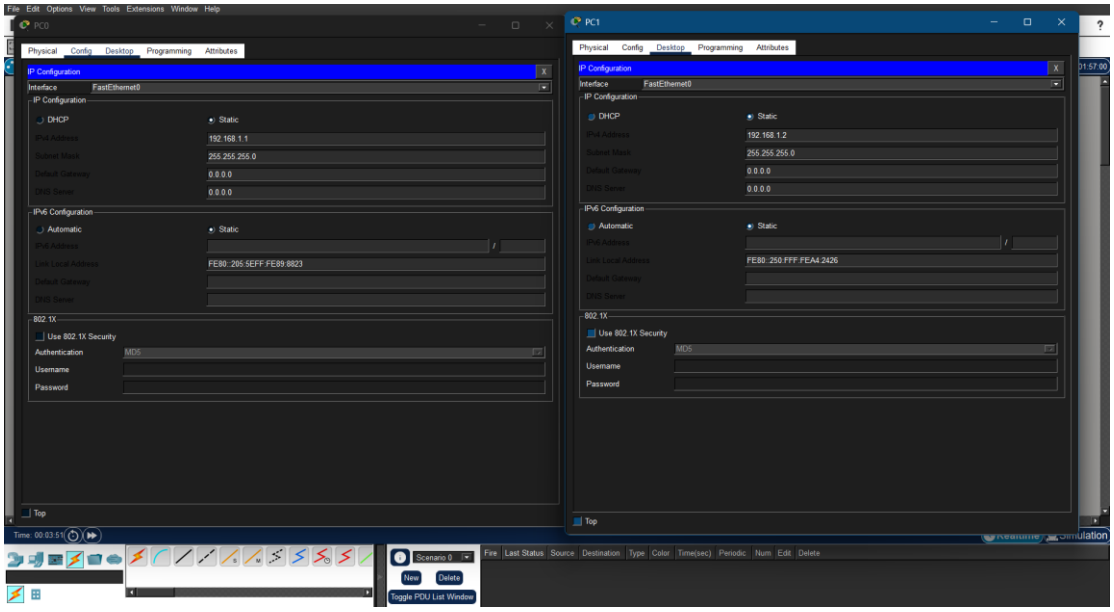
ARP Request:

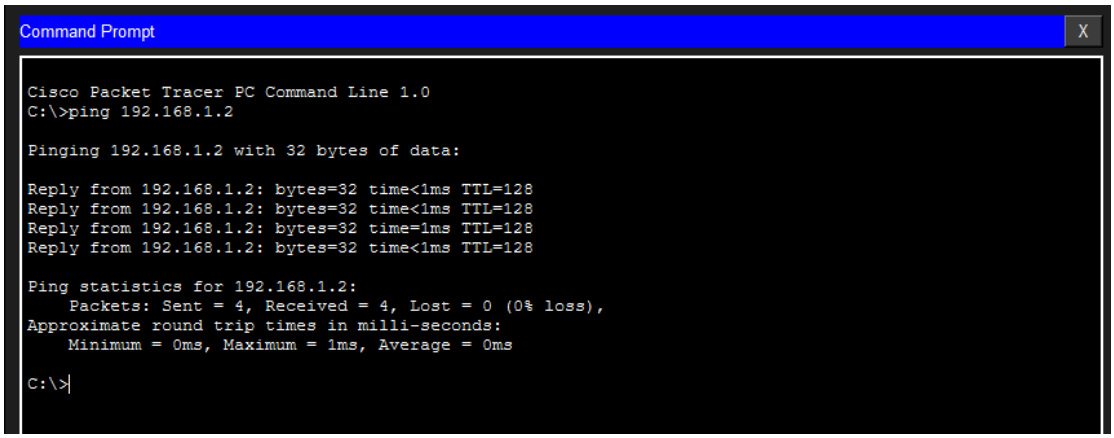


ARP Response:

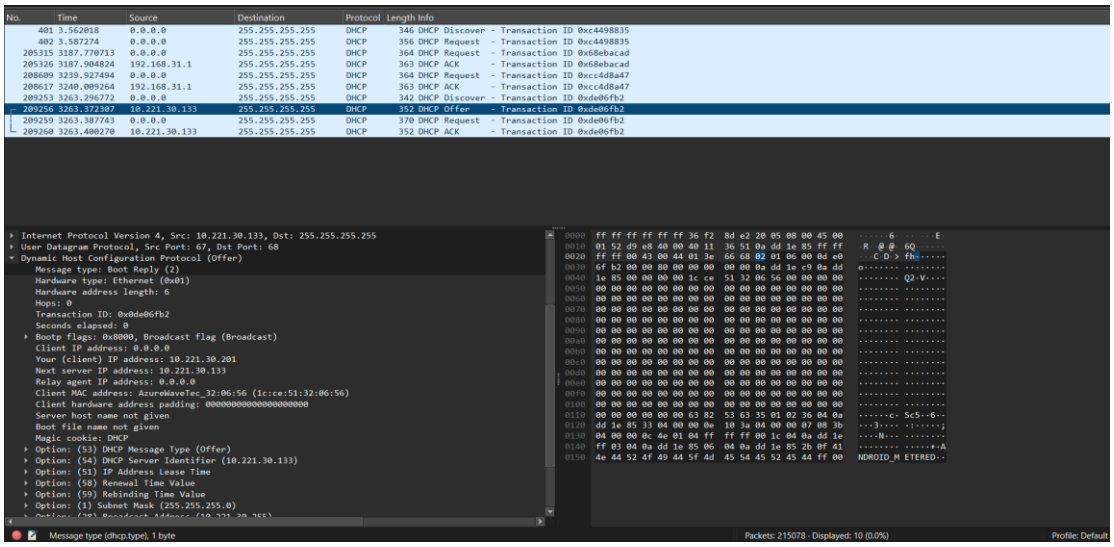


3)


PC0
PC1
IP Configuration
Interface FastEthernet0
DHCP
Static
192.168.1.1
255.255.255.0
192.168.1.2
IPv6 Configuration
802.1X
Use 802.1X Security
Authentication
Username
Password


Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>

4)


No. Time Source Destination Protocol Length Info
401 3.562018 0.0.0.0 255.255.255.255 DHCP 346 DHCP Discover - Transaction ID 0xc4498835
402 3.587274 0.0.0.0 255.255.255.255 DHCP 356 DHCP Request - Transaction ID 0xc4498835
205315 3187.770713 0.0.0.0 255.255.255.255 DHCP 364 DHCP Request - Transaction ID 0x68ebacad
205326 3187.904824 192.168.31.1 255.255.255.255 DHCP 363 DHCP ACK - Transaction ID 0x68ebacad
208609 3239.927494 0.0.0.0 255.255.255.255 DHCP 364 DHCP Request - Transaction ID 0xcc4d8a47
208617 3240.009264 192.168.31.1 255.255.255.255 DHCP 363 DHCP ACK - Transaction ID 0xcc4d8a47
209253 3263.296772 0.0.0.0 255.255.255.255 DHCP 342 DHCP Discover - Transaction ID 0xde06fb2
209256 3263.372307 10.221.30.133 255.255.255.255 DHCP 352 DHCP Offer - Transaction ID 0xde06fb2
209259 3263.387743 0.0.0.0 255.255.255.255 DHCP 370 DHCP Request - Transaction ID 0xde06fb2
209260 3263.400270 10.221.30.133 255.255.255.255 DHCP 352 DHCP ACK - Transaction ID 0xde06fb2
Internet Protocol Version 4, Src: 10.221.30.133, Dst: 255.255.255.255
User Datagram Protocol, Src Port: 67, Dst Port: 68
Dynamic Host Configuration Protocol (Offer)
Message type: Boot Reply (2)
Hardware type: Ethernet (0x01)
Hardware address length: 6
Hops: 0
Transaction ID: 0x0de06fb2
Seconds elapsed: 0
Bootp flags: 0x8000, Broadcast flag (Broadcast)
Client IP address: 0.0.0.0
Your (client) IP address: 10.221.30.201
Next server IP address: 10.221.30.133
Relay agent IP address: 0.0.0.0
Client MAC address: AzureWaveTec_32:06:56 (1c:ce:51:32:06:56)
Client hardware address padding: 00000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP
Option: (53) DHCP Message Type (Offer)
Option: (54) DHCP Server Identifier (10.221.30.133)
Option: (51) IP Address Lease Time
Option: (58) Renewal Time Value
Option: (59) Rebinding Time Value
Option: (1) Subnet Mask (255.255.255.0)
Message type (dhcp.type), 1 byte
Packets: 215078 · Displayed: 10 (0.0%)
Profile: Default

5)

Step 1: Determine New Subnet Mask

- **Given:** /24
- **Need:** 4 subnets
- **Formula:**

2^n=4

n=2

Borrow 2 bits

New Subnet Mask:

/26 → 255.255.255.192

Step 2: Subnet Details

Each subnet has:

- Total IPs = 64
- Usable hosts = 62

Step 3: Subnet Ranges

Subnet	Network Address	Usable Range	Broadcast
1	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
2	192.168.1.64/26	192.168.1.65 – 192.168.1.126	192.168.1.127
3	192.168.1.128/26	192.168.1.129 – 192.168.1.190	192.168.1.191
4	192.168.1.192/26	192.168.1.193 – 192.168.1.254	192.168.1.255

Step 4: Packet Tracer Assignment (Example)

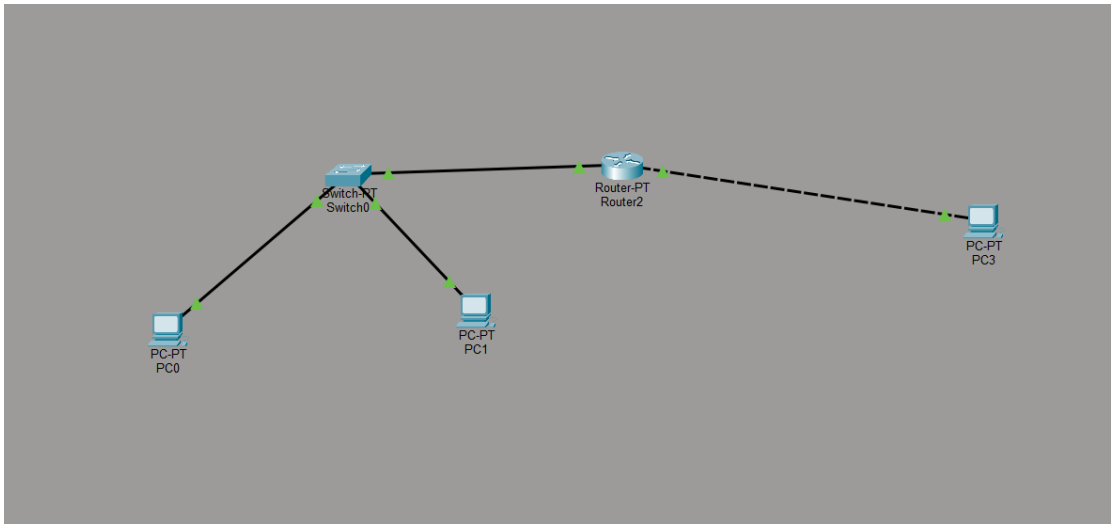
- **Subnet 1:**
  - PC1 → 192.168.1.1
  - PC2 → 192.168.1.2
- **Subnet 2:**
  - PC3 → 192.168.1.65

Devices in same subnet → **Ping works**
Different subnet → **Need router**

6)

IP Address	Class	Default Subnet Mask
10.1.1.1	Class A	255.0.0.0 (/8)
172.16.5.10	Class B	255.255.0.0 (/16)
192.168.1.5	Class C	255.255.255.0 (/24)

7)




```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.1     YES manual up                    up
FastEthernet1/0        10.0.0.1        YES manual up                    up
Serial2/0              unassigned      YES unset  administratively down down
Serial3/0              unassigned      YES unset  administratively down down
FastEthernet4/0        unassigned      YES unset  administratively down down
FastEthernet5/0        unassigned      YES unset  administratively down down
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 10.0.0.1:45       192.168.1.2:45     10.0.0.2:45        10.0.0.2:45
icmp 10.0.0.1:46       192.168.1.2:46     10.0.0.2:46        10.0.0.2:46
icmp 10.0.0.1:47       192.168.1.2:47     10.0.0.2:47        10.0.0.2:47
icmp 10.0.0.1:48       192.168.1.2:48     10.0.0.2:48        10.0.0.2:48

Router#
```

The show ***ip nat translations*** command shows that the private IP (192.168.1.2) is translated to the public IP (10.0.0.1) with different port numbers, confirming that NAT overload is working correctly and enabling successful communication with the external network.